

5-Mark Questions & Answers — MPCA Module 4: Memory & Peripherals Interfacing

Q1. Explain 8255 PPI internal block diagram, operating modes, and interface with 8086.

- Block diagram (Data buffer, Read/Write logic, Group A & B controls, Ports A, B, C).
 - Mode 0, Mode 1, Mode 2, and BSR bit set/reset format.
 - Pin connection with 8086 Data lines **D₀-D₇**, Address lines **A₁-A₂**, and chip select logic.
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Q2. Explain 8257 DMAC block diagram, register set, and DMA transfer modes.

- Architecture of 4 channels, Address/Count registers, and Priority Resolver.
 - Single, Block (Burst), Demand, and Cascade transfer modes.
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Q3. Explain 8259 PIC internal architecture, interrupt sequence, and priority modes.

- Detailed breakdown of IRR, ISR, IMR, Priority Resolver, and Control logic.
 - Step-by-step two-/INTA pulse handshake and vector delivery to 8086 IVT.
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